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ARTool: R Package for the Aligned Rank Transform for Nonparametric Factorial ANOVAs



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ARTool is an R package implementing the Aligned Rank Transform for conducting nonparametric analyses of variance on factorial models. This implementation is based on the ART procedure as used in the original implementation of [ARTool](#) by Wobbrock et al.

The package automates the Aligning-and-Ranking process using the `art` function. It also automates the process of running a series of ANOVAs on the transformed data and extracting the results of interest. It supports traditional ANOVA models (fit using `lm`), repeated measures ANOVAs (fit using `aov`), and mixed effects models (fit using `lmer`); the model used is determined by the formula passed to `art`.

Note: The documentation of this package assumes some level of familiarity with when and why you may want to use the aligned rank transform; the [ARTool page](#) provides a more in-depth (and highly approachable) introduction to the aligned rank transform and the motivation for its use.

Installation

You can install the latest released version from CRAN with this R command:

```
install.packages("ARTool")
```



Or, you can install the latest development version from GitHub with these R commands:

```
install.packages("devtools")  
devtools::install_github("mjskay/ARTool")
```



Example

The general approach to using ART is to transform your data using `art`, verify the ART procedure is appropriate to the dataset using `summary`, and then run an ANOVA on the transformed data using `anova`.

First, let us load some example data:

```
library(ARTool)
data(Higgins1990Table5, package = "ARTool")
```



Higgins1990Table5 is a data frame from an experiment in which the effects of `Moisture` and `Fertilizer` on `DryMatter` in peat pots was tested. Four pots were placed on each `Tray`, with `Moisture` varied between `Tray`s and `Fertilizer` varied within `Tray`s. We can see the basic structure of the data:

```
str(Higgins1990Table5)
```



```
## 'data.frame': 48 obs. of 4 variables:
## $ Tray : Factor w/ 12 levels "t1","t2","t3",...: 1 1 1 1 2 2 2 2 3 3 ...
## $ Moisture : Factor w/ 4 levels "m1","m2","m3",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ Fertilizer: Factor w/ 4 levels "f1","f2","f3",...: 1 2 3 4 1 2 3 4 1 2 ...
## $ DryMatter : num 3.3 4.3 4.5 5.8 4 4.1 6.5 7.3 1.9 3.8 ...
```



```
head(Higgins1990Table5, n=8)
```



```
## Tray Moisture Fertilizer DryMatter
## 1 t1 m1 f1 3.3
## 2 t1 m1 f2 4.3
## 3 t1 m1 f3 4.5
## 4 t1 m1 f4 5.8
## 5 t2 m1 f1 4.0
## 6 t2 m1 f2 4.1
## 7 t2 m1 f3 6.5
## 8 t2 m1 f4 7.3
```



Step 1: Transform the data

To analyze this data using the aligned rank transform, we first transform the data using `art`. We specify the response variable (`DryMatter`), the fixed effects and all of their interactions (`Moisture*Fertilizer`, or equivalently `Moisture + Fertilizer + Moisture:Fertilizer`), and any grouping terms if present (here, `(1|Tray)`).

While `(1|Tray)` has no effect on the results of the aligned rank transformation, it will be used by `anova` to determine the type of model to run: when grouping terms are present, mixed effects models are run using `lmer`. If you wish to use a repeated measures ANOVA instead of a mixed effects model, you can use an `Error` term instead (see below for an example of this). If you do not have repeated measures, do not include any grouping terms or error terms.

```
m <- art(DryMatter ~ Moisture*Fertilizer + (1|Tray), data=Higgins1990Table5)
```



Step 2: Verify appropriateness of ART

To verify that the ART procedure was correctly applied and is appropriate for this dataset, we can look at the output of `summary` :

```
summary(m)
```

```
## Aligned Rank Transform of Factorial Model
##
## Call:
## art(formula = DryMatter ~ Moisture * Fertilizer + (1 | Tray),
##      data = Higgins1990Table5)
##
## Column sums of aligned responses (should all be ~0):
##           Moisture           Fertilizer Moisture:Fertilizer
##                0                0                0
##
## F values of ANOVAs on aligned responses not of interest (should all be ~0):
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
```

We see that the columns sums of aligned responses and the F values of ANOVAs on aligned responses not of interest are all ~0, indicating that the alignment correctly “stripped out” effects not of interest. Thus, we can apply the ANOVA on the transformed data.

Step 3: Run the ANOVA

ARTool automatically selects the model to be used for the ANOVA. Because we have included a grouping term, `(1|Tray)`, ARTool will fit mixed effects models using `lmer` and run the ANOVAs on them:

```
anova(m)
```

```
## Analysis of Variance of Aligned Rank Transformed Data
##
## Table Type: Analysis of Deviance Table (Type III Wald F tests with Kenward-Roger df)
## Model: Mixed Effects (lmer)
## Response: art(DryMatter)
##
##              F Df Df.res      Pr(>F)
## 1 Moisture      23.833  3      8 0.00024199 ***
## 2 Fertilizer    122.402  3     24 1.1124e-14 ***
## 3 Moisture:Fertilizer  5.118  9     24 0.00064665 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Alternative model: Repeated Measures ANOVA

This particular study could also be analyzed using a repeated measures ANOVA, yielding the same results (note that repeated measures ANOVAs and mixed effects models will not always yield the same results). To instead run a repeated measures ANOVA, add an `Error` term to the model as you might for a call to `aov` :

```
m <- art(DryMatter ~ Moisture*Fertilizer + Error(Tray), data=Higgins1990Table5)
anova(m)
```



```
## Analysis of Variance of Aligned Rank Transformed Data
##
## Table Type: Repeated Measures Analysis of Variance Table (Type I)
## Model: Repeated Measures (aov)
## Response: art(DryMatter)
##
##
##          Error Df Df.res F value    Pr(>F)
## 1 Moisture      Tray  3      8  23.833 0.00024199 ***
## 2 Fertilizer    Withn  3     24 122.402 1.1124e-14 ***
## 3 Moisture:Fertilizer Withn  9     24   5.118 0.00064665 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Contrast tests

For an example of how to run contrast tests on an `art` model, see `vignette("art-contrasts")`.

Problems

Should you encounter any bugs in this package, please file it [here](#) with minimal code to reproduce the issue.

Citations

Kay, M., Elkin, L. A., Higgins, J. J., and Wobbrock, J. O. (2025). *ARTool: Aligned Rank Transform for Nonparametric Factorial ANOVAs*. R package version 0.11.2, <https://github.com/mjskay/ARTool>. DOI: [10.5281/zenodo.594511](https://doi.org/10.5281/zenodo.594511).

For the ART procedure used by `art()` and `anova.art()`, cite:

Wobbrock, J. O., Findlater, L., Gergle, D., and Higgins, J. J. (2011). The Aligned Rank Transform for Nonparametric Factorial Analyses Using Only ANOVA Procedures. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2011)*. Vancouver, British Columbia (May 7-12, 2011). New York: ACM Press, pp. 143-146. <https://depts.washington.edu/accelab/proj/art/>. DOI: [10.1145/1978942.1978963](https://doi.org/10.1145/1978942.1978963).

For the ART-C contrast testing procedure used by `art.con()` and `artlm.con()`, cite:

Elkin, L. A., Kay, M., Higgins, J. J., and Wobbrock, J. O. (2021). An Aligned Rank Transform Procedure for Multifactor Contrast Tests. *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2021)*. Virtual Event (October 10-14, 2021). New York: ACM Press, pp. 754-768. DOI: [10.1145/3472749.3474784](https://doi.org/10.1145/3472749.3474784)

Releases 12

 **ARTool 0.11.2** Latest
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Contributors 2

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Languages

 **R** 100.0%